

Unlocking the Potential of 2D Ferroelectrics with Machine Learning: A Review

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Abstract – Two-dimensional (2D) ferroelectrics offer switchable polarisation in atomically thin films, enabling ultra-compact, energy-efficient sensors, memories, and energy harvesters. However, research on 2D ferroelectric devices is limited: the growth of monolayers via molecular beam epitaxy (MBE), chemical vapour deposition (CVD), and related techniques remains challenging, while conventional modelling methods such as density functional theory (DFT) and molecular dynamics (MD) are computationally intensive. Machine learning (ML) has emerged as a powerful tool to bridge this gap by identifying patterns in existing data. ML models can rapidly predict key ferroelectric properties including Curie temperature, band gap, polarisation, and also screen large chemical spaces for promising materials. In this review, we summarise the fundamentals of ferroelectricity and finite-size effects in 2D crystals, before introducing ML strategies tailored to this domain. We highlight representative case studies: classification of polar versus non-polar structures, regression of exfoliation energies to identify easily exfoliable films, and deep learning analysis of atomic-resolution scanning transmission electron microscopy (STEM) images to map polarisation with near real-time speed. Finally, we discuss the advantages and limitations of ML approach in 2D ferroelectric research, and offer an outlook. By combining physics and data science, ML approaches promise to significantly accelerate the discovery and understanding of 2D ferroelectric materials.

I INTRODUCTION

Two-dimensional ferroelectrics (2D FEs) are atomically thin crystals that combine the unique physics of ultra-thin materials with switchable spontaneous polarisation. Examples include monolayer In_2Se_3 and thin films of BaTiO_3 [1, 2], which promise compact energy-efficient sensors, memories, and energy harvesters [3, 4, 5, 6]. While being a hotspot for experimental and theoretical works, the diverse underlying mechanisms, as well as copious material properties, are still awaiting further exploration [7].

On the one hand, experimental synthesis of monolayer or few-layer ferroelectrics often requires intricate growth techniques such as molecular beam epitaxy or chemical vapour deposition [8, 9]. On the other hand, theoretical

predictions of 2D FE properties rely heavily on first-principles calculations. Traditional computational methods, such as density functional theory (DFT) and molecular dynamics, are highly accurate but computationally intensive [10, 11].

In the face of these challenges, machine learning (ML) has emerged as a powerful tool. By identifying patterns in both experimental data and computational results, ML models can efficiently screen novel candidates and estimate key properties such as Curie temperature T_c [12], band gaps, electrical breakdown limits [13] and polarisation distributions [14] in 2D FE systems with sufficient accuracy as well as helping with experiments by finding exfoliation energy [15] and mechanical properties [16] of layered structures.

This review first recaps the physics of 2D ferroelectrics, highlighting essential concepts like soft-mode theory and polarization mechanisms. We then outline ML strategies for 2D FE science (data sources, feature selection and algorithms choices). Selected case studies illustrate how ML can accelerate 2D ferroelectric research by linking prediction, experimental validation, and result analysis. Finally, we discuss the advantages and limitations of ML in this field and argue that combining physics insight with ML can greatly accelerate understanding of 2D FEs.

II THEORETICAL BACKGROUND

A General Ferroelectricity

Ferroelectricity is exhibited by materials with spontaneous polarisation and is reversible through the application of an electric field [17].

The microscopic mechanism of ferroelectric polarisation involves competing interactions. Conventionally, soft-mode theory gives a good explanation of displacive FE polarisation [18]. The soft mode theory explains ferroelectric phase transitions in terms of lattice vibrations (phonons), specifically a transverse phonon (TO) whose frequency decreases (“softens”) as the temperature approaches T_c , ($\omega^2(T) \propto (T - T_c)$). This indicates that the lattice can lower its free energy by

distorting along the corresponding phonon eigenvector. Once $T < T_c$, the optic phonon mode becomes unstable and the system spontaneously distorts into a lower symmetry, polar structure which has ferroelectricity with a non-zero spontaneous polarisation P .

Macroscopically, ferroelectric phase transitions and domain energetics are often modelled by Landau–Ginzburg–Devonshire (LGD) theory [19, 20]. In simplest form, the free energy as a function of polarisation $\mathbf{P}$ and temperature T is written:

$$F[P, T] = \frac{1}{2}\alpha(T - T_c)P^2 + \frac{1}{4}\beta P^4 + \frac{1}{6}\gamma P^6 + \frac{1}{2}\kappa(\nabla P)^2 - EP, \quad (1)$$

where α, β, γ are material-specific coefficients, κ accounts for domain wall energy, and E is the external electric field.

B Polarization Mechanisms in 2D

In atomically thin crystals, ferroelectricity can arise by several distinct mechanisms. These examples (Table 1) illustrate that 2D ferroelectricity may not involve the classic perovskite mechanism; unique 2D geometries and van der Waals physics open new routes to polar behaviour.

C Limitations on Ferroelectricity in 2D

Reducing the dimension generally suppresses ferroelectricity. For example, bound surface charge $\sigma_b = P \cdot \hat{n}$ produces a depolarising field $E_d \propto P/t$ that grows sharply with shrinking thickness t [26]. When the electrostatic cost is higher than the free-energy gain, the polarised state becomes unstable.

Conversely, numerous studies have demonstrated that the Curie temperature (T_c), the temperature required to maintain the ferroelectric phase, decreases as film thickness reduces [12]. A study found that a four-layer BaTiO₃ film has $T_c \approx 70K$ [27], which is much lower than bulk – so many candidate 2D FEs may not remain ferroelectric at room temperature. Electronic screening and leakage, gradient energy and mechanism dependence are other possible limitations which we will not discuss in detail here.

D Practical Limits of Conventional Theory

As noted earlier, LGD theory represents a ferroelectric monolayer with a few temperature-dependent coefficients, $\alpha, \beta, \gamma, \kappa$, which determine the double-well depth, the domain wall energy, and the critical thickness. Obtaining these coefficients still requires first-principles calculations that do not scale well: each composition, layer number, and strain state needs dozens of

frozen-phonon or density-functional-perturbation theory (DFPT) calculations, followed by finite-temperature sampling to check T_c . Screening 1,000 possible two-dimensional ferroelectrics therefore requires on the order of 10^5 such calculations – up to years of supercomputer time. Experiments require even more time and effort, so they tend to focus on structures that have already been tested by first-principles studies.

Machine learning (ML) offers a way to fill this gap by accelerating the search while using far fewer computational and experimental resources. The next section explains in detail how ML can be applied to two-dimensional ferroelectrics and why it is effective.

III MACHINE LEARNING APPROACHES IN 2D FE SCIENCE

A An Introduction to Machine Learning

Machine learning addresses the question of how to build computers that improve automatically through experience [28]. It has been a rapid developing field in the past two decades. Various machine learning algorithms are being classified by their learning patterns into supervised learning, unsupervised learning, reinforcement learning and other minor divisions. In supervised learning, a model learns a mapping $f : \mathbf{X} \rightarrow Y$ from labelled pairs $(\mathbf{X}, Y)$ and thus useful in interpreting LGD theory; in unsupervised learning it extracts patterns directly from unlabelled data, so it can discover new ferroelectrics by extracting data from material databases; reinforcement learning optimises sequential decision-making, sometimes guiding valuable computational or experimental resources towards the most promising regions of research.

Ferroelectrics research typically uses supervised and unsupervised learning. Common algorithm choices are shown in Figure 1.

B Applying ML to 2D FE Studies

Applying ML to 2D ferroelectrics involves several stages: assembling data (from experiments or simulations), selecting descriptors (features), choosing models, and interpreting results. Below we outline key considerations, demonstrating how they relate to physical insight.

B.1 Data Collection

Data for 2D ferroelectrics come from both experiment and computation. There have been many open source databases with material parameters and properties.

Mechanism	Physical origin in 2D	Representative materials
Intrinsic structural	Off-centre of ions in puckered/hinge lattices	SnTe [21], SnSe, In ₂ Se ₃ [22]
Layer-asymmetric dipole	Inequivalent sublayers (chalcogen/metal slabs)	CuInP ₂ S ₆ , CuBiP ₂ S ₆ [23]
Sliding/stacking (vdW)	Lateral shift or twist of two neutral layers	Bilayer h-BN [24], twisted SnSe ₂
Moiré ferroelectricity	Layer-registry reconstruction in twisted bilayers	Twisted WSe ₂ , MoS ₂ , h-BN [25]

Table 1: Mechanisms enabling ferroelectricity in atomically thin crystals.

#	Database	Primary Data Source(s)	Scope (2025)
1	2DMatPedia [30]	DFT (VASP, PBE)	~6 500 monolayers
2	C2DB [31]	DFT (GPAW) + GW/BSE	~4 000 relaxed sheets
3	JARVIS (2D subset) [32]	DFT (optB88-vdW, HSE); MD/FF	~1 000 entries
4	Materials Cloud 2D Crystals [33, 34]	DFT (“computational exfoliation”)	2 004 easily exfoliable compounds
5	The Materials Project [35]	DFT	–
6	NOMAD Repository [36]	Multi-code DFT	>4 680 monolayers
7	2DMat.ChemDX	Experimental (RHEED, Raman, PL)	$\mathcal{O}(10^2)$ samples

Table 2: Representative open-access databases containing two-dimensional materials.

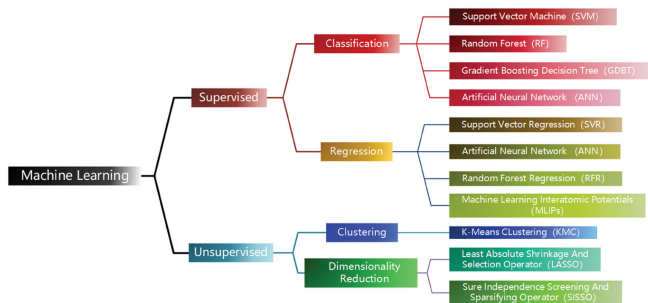


Figure 1: A few popular machine learning algorithms in material property studies classified into supervised and unsupervised learning. Figure adopted from [29].

Here is a collection of popular open source material database for 2D materials in Table 2.

The relevant physical scales are also crucial. Polarisation phenomena span atomic scales (angstroms, THz lattice dynamics) up to macroscopic scales (domain patterns, micrometres, and MHz). Meanwhile, computational methods cover wide ranges: DFT treats unit cells, molecular dynamics spans nanoseconds, and phase-field models cover micrometres of domain structure. Figure 2 illustrates how the time/length scales of polarisation mechanisms compare to those of typical modelling methods.

For example, if we are interested in the temperature-dependent thermodynamics or domain-wall kinetics of a material, which may contain the dynamics of $\mathcal{O}(10^6)$ atoms in several nanoseconds of time. Now the quantum mechanical features are less important and we can treat the atoms classically through the Force-field method (part of molecular dynamics [11]) which describes the

dependence of the energy of a system on the coordinates of its particles [37].

B.2 Feature Engineering and Descriptors

For crystalline materials, inputs to ML models typically include atomic-level descriptors (atomic numbers, coordination, bond lengths, etc.), crystal structural features, or orbital populations. Physically motivated descriptors are crucial. For example, Ma et al. constructed orbital-based features (such as ionic radii and electronegativities) to train a classifier for 2D ferroelectrics [40]. In general, one should choose descriptors that reflect the physics of interest: for polarization or stability, features like atomic displacements, layer thickness, or bond polarity can be important. While most of the existing summarisation of potential descriptors are for general bulk FEs [41], this can still be inspirational to the 2D systems.

It is also worth noting that trying out and identifying good descriptors through the training process can bring insights to the deep-lying mechanisms. In a work carried out by Wang et al. [42], the descriptor: *effective* van der Waals radius was noticed to be efficient in modelling homobilayer sliding FEs. It provides another possible direction for further investigation.

B.3 Choice of Algorithms

Predicting the properties of two-dimensional (2D) ferroelectric materials with machine learning requires choosing algorithms that suit the nature of the input data. Different data sources – DFT simulations, experimental measurements, and classical force-field simulations – have varying data volumes, noise levels,

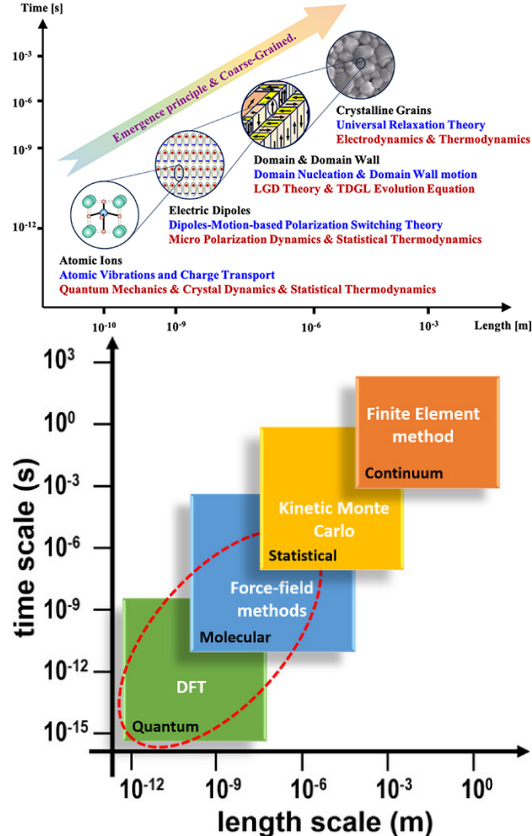


Figure 2: A comparison of the time/length scales of the multiscale nature of polarisation with the scales of popular computational methods. Figures adopted from [38, 39]

and feature types. Here, we will only recommend a few suitable ML algorithms for each data type as listed in Table 3.

DFT data are highly accurate but often limited due to computational costs, so ML models must handle small datasets with many descriptors.

Force-field simulations, such as molecular dynamics, produce large but less accurate datasets which often with varying conditions or data in time series. ML methods should utilise this volume to capture complex patterns.

Experimental data, such as polarisation or Curie temperature, are reliable but scarce and noisy. As experiments usually cover limited conditions, algorithms must avoid over-fitting and suit for small, noisy datasets with interpretability.

In practice, exact choice of algorithm can be made by trying out a list of suitable algorithms. Also, studies sometimes use a combination of different algorithms. For example, a recent work of discovering new ferroelectric materials used stages of ML: initial stability filters with ensemble classifiers, then a random forest regression to

predict polarisation magnitude [47].

Details of model training and optimisation are beyond this review, but readers can explore relevant ML literature for deeper insights [48, 49].

C Existing Researches

ML has already been employed to solve various challenges in 2D FE materials. Some focus on discovering new material, while others build property predictors or approximation (surrogate) models to replace costly simulations. Notably, ML has not only been used with computational data, but also experimental results, showing that it's highly generalizable.

We will dive into a few of these examples in the [CASE STUDY](#) section.

IV CASE STUDIES

To illustrate the above ideas, we highlight a three-stage workflow where ML has advanced 2D ferroelectric science. Through these case studies, we see how ML methods, from predictive modelling to regression and deep neural networks, provide a powerful toolkit that spans theory, materials synthesis, and characterisation, greatly accelerating discovery and understanding in 2D ferroelectrics.

A ML-Guided Theoretical Verification

Ma et al.[40] investigated an extensive family of 2D bimetallic phosphates $M_I M_{II} P_2 X_6$ (≈ 2964 compounds) in search of ferroelectricity. The task was framed as a binary classification: predict whether a given 2D compound is polar (ferroelectric) or non-polar. Using high-throughput first-principles data, they developed a classifier with new electronic orbital-based descriptors as features. It was later found that coinage metal elements and IIIA/IVA metal elements, due to their full d-orbitals, do not bond octahedrally with six chalcogen atoms as expected. They rather bond in a lower-symmetry way, resulting in spontaneous symmetry breaking and thus out-of-plane polarisation. This finding is consistent with the improvement to classification model brought by introducing new orbital-based descriptor.

Several ML algorithms (support vector classifiers, decision trees, random forests, AdaBoost, etc.) were tested; a gradient-boosted trees model ultimately yielded the best performance (77.2% accuracy, 88.3% AUC (the area under the receiver operating characteristic curve, a graphical plot that illustrates the performance of a binary classifier model)) This model could successfully discern polar vs. non-polar structures across the chemical space. Crucially, the ML model identified 60

Table 3: Summary of Recommended Algorithms for Different Data Sources

Data Source	Recommended Algorithms	Reasoning	Notes
DFT calculations	- Random Forest (RF) [43] - Support Vector Machine (SVM) [44] - Gradient Boosted Trees (XGBoost [45]) - Neural Networks (NN) - Gaussian Process Regression (GPR) [46]	- RF: useful for small to medium datasets, interpretable - SVM: for complex but small datasets - Boosting: most accurate if tuned well - NN: for large datasets, flexible - GPR: best when data is very small and uncertainty is needed	- RF and GPR preferred for moderate-size DFT data - NNs useful if thousands of DFT points or graph-based input (crystal structures)
Force-field simulations	- Neural Networks (Deep Learning) - Random Forest (RF) - Gradient Boosted Trees (XGBoost) - Gaussian Process Regression (GPR)	- NN: best for large, complex, high-dimensional simulation data - RF/Boosting: fast, interpretable if data can be broken down to each parameter - GPR: good for surrogate modelling when the number of simulations is moderate	- NNs excellent when simulation data are abundant - RF/Boosting practical for fast prediction from structured outputs (e.g., polarisation vs temperature curves)
Experimental measurements	- Random Forest (RF) - Support Vector Machine (SVM) - Gaussian Process Regression (GPR)	- RF: handles noise well, interpretable - SVM: good for small datasets - GPR: provides uncertainty estimates, ideal for noisy small datasets	- Interpretability is important for experiments; RF usually best first choice - Deep learning (NN) only if data is relatively large (hundreds of samples)

previously unrecognised stable ferroelectric 2D materials, of which 16 were ferroelectric metals (possessing spontaneous polarisation and metallic conductivity). The workflow and comparison of algorithms are shown in Figure 3. These findings greatly expanded the family of 2D ferroelectrics.

This study shows how ML can rapidly explore thousands of compounds to pinpoint rare phenomena – in this case discovering dozens of 2D ferroelectric materials. And these structure or property predictions can then be sent off for experimental verification.

B ML-Prioritised Synthesis

Wan et al. [15] addressed the challenge of identifying 2D materials that can be synthesised via exfoliation by predicting exfoliation energies using ML. They adapted a dataset of 4000 2D materials (from the 2DMatPedia database) with DFT-calculated exfoliation energies and engineered 12 descriptors: 6 related to van der Waals bonding, 5 to electrical properties, and 1 to bulk stability (decomposition energy). Several regression models (support vector regression, linear regression, decision/ensemble trees) were tested, with an ensemble of regression trees performing best ($R^2 \approx 0.89$, mean absolute error ≈ 30 meV/atom), as shown in Figure 4. This accuracy is sufficient to distinguish exfoliable materials, with the threshold typically set around 150 meV/atom. The model effectively filters thousands

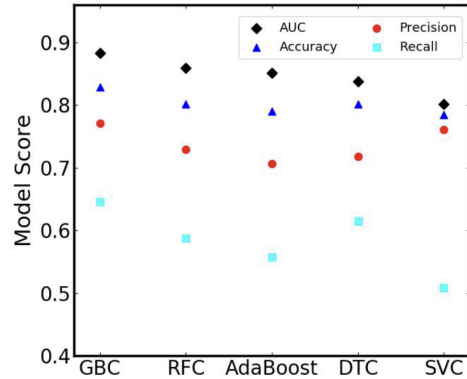
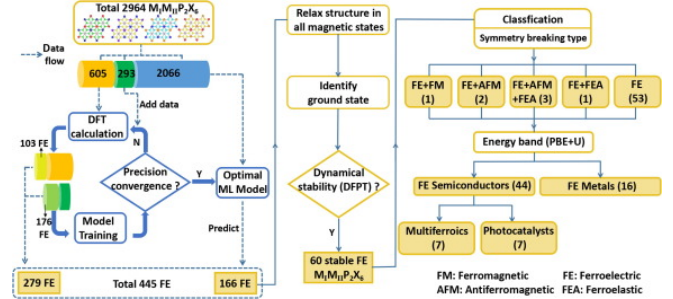


Figure 3: Figures adopted from fig.2 and fig.S2 in “Large Family of Two-Dimensional Ferroelectric Metals Discovered via Machine Learning” by Ma et al. [40]

of candidates, flagging those below this threshold as synthesisable.

ML has also been applied to chemical vapour deposition (CVD) [50, 51], where gaseous or vapour-phase materials react at gas-phase or gas-solid interfaces to form solid deposits [52]. Although not yet used specifically for 2D ferroelectrics, it holds promise for this area with appropriate generalisation and adaptation.

As a result, one can prioritise 2D ferroelectric candidates that are not only theoretically interesting but also experimentally obtainable. This ML-driven exfoliation predictor bridges the gap between theory and practice by ensuring that the proposed 2D ferroelectrics have a viable route to fabrication.

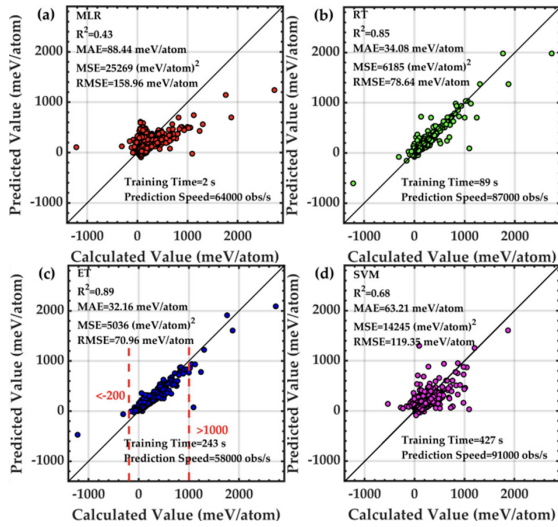


Figure 4: Scatter plot of the exfoliation energy values in the data set against the ML predicted values for the test set using different ML model. [15]

C ML-Assisted Experimental Characterisation

Atomic resolution STEM enables polarisation mapping by detecting sub-ångström displacements of atomic columns, particularly between cations and anions in perovskite-like structures [53, 54]. This displacement vector defines the polarisation P and mapping P across the sample unveils domain patterns.

Nelson et al. trained a deep convolutional neural network (DCNN) with a labelled dataset of 8000 image patches (80×80 pixels) to predict polarisation distributions directly from STEM images[14]. The method reproduced known domain features and phase transitions induced by Sm doping, where polarisation patterns evolve from rhombohedral (pure) to monoclinic ($\sim 7 - 10\%$) and eventually non-polar phases ($\sim 20\%$) (see Figure 5).

ML predictions were consistent with displacement-derived values, capturing both direction and magnitude of P . A key advantage was speed: the trained DCNN could infer polarisation maps for 256×256 pixels images in just 1–2 seconds. However, inaccuracies near defects highlighted the continued need for physical interpretation, that local symmetry breaking, defects, and lattice distortions still govern polarisation behaviour.

This case highlights how deep learning can accelerate experimental workflows by rapidly converting imaging data into quantitative polarisation information, enabling real-time observation of ferroelectric domain dynamics.

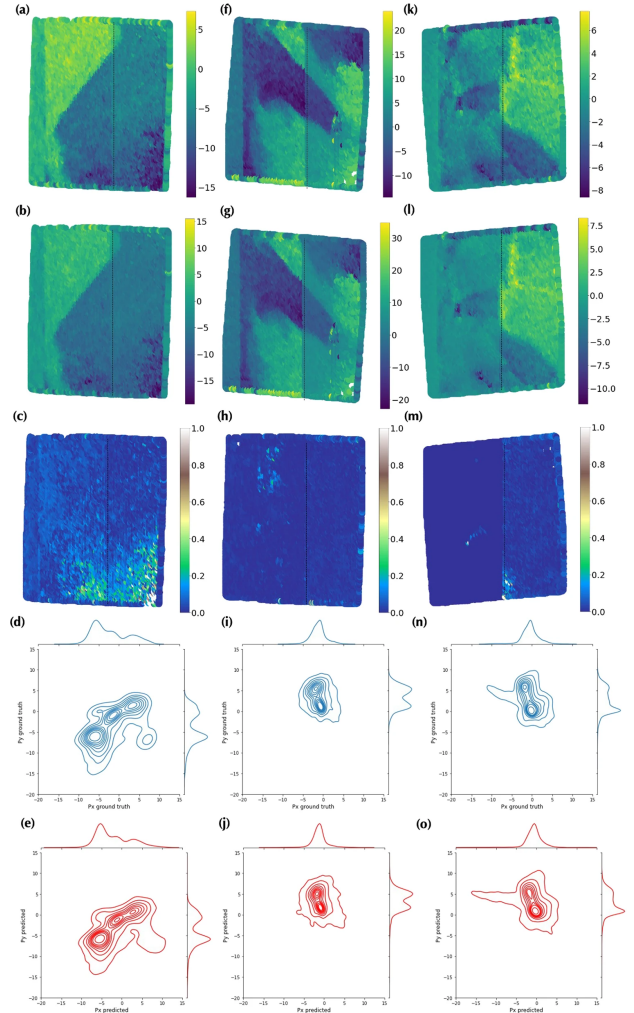


Figure 5: Polarisation maps (via NC) showing ground truth (top), predictions (middle), and 2D histograms of P_x and P_y components (bottom rows) are provided for 0% (a–e), 7% (f–j), and 10% (k–o) Sm-doped BFO. Predictions are from three networks trained on two-thirds of sub-images per concentration and tested on the remainder. Vertical lines in plots indicate train-test splits.[14]

More studies further demonstrate the wide-ranging use of ML in 2D ferroelectrics. We will only mention a few here. For example, Hao et al. used ML classification on high-throughput DFT data to discover new 2D multiferroic semiconductors [55]. Wang et al. applied big-data analysis and ML to screen bilayer heterostructures, identifying 25 “sliding” ferroelectric bilayers with strong out-of-plane polarisation and low sliding barriers [42]. Similarly, Wu et al. developed a deep neural network interatomic potential to simulate structural phase transitions and domain switching in the monolayer In_2Se_3 [56]. Together, these examples show the broad applicability of ML (from materials screening to dynamic modelling) in advancing 2D ferroelectric research. This diversity of approaches sets the stage for the final considerations and outlook presented in Section V.

V FINAL CONSIDERATIONS AND OUTLOOK

As we discussed, machine learning has already shown clear benefits for 2D ferroelectrics. Once trained, ML models can greatly reduce the cost of predictive modelling, materials discovery, and even guiding synthesis.

However, ML also has limitations. It requires large amounts of high-quality training data, but in the relatively young 2D ferroelectric field, such data are still limited, which reduces model accuracy. The predictive performance of ML depends heavily on the training data: while a high accuracy (e.g. $R^2 \approx 0.9$) is impressive but not universal. Models can make large errors when applied to configurations that are different from the training data.

Interpretability is another concern. Many models behave as “black boxes” whose internal decision making process is hard to trace. This can limit trust and physical insight. Efforts like feature importance in random forests or attention maps in neural networks help, but so far only offer limited insight. Thus, it is important to carefully check ML predictions in 2D ferroelectrics with physical reasoning or conventional methods to ensure that the results are meaningful.

Looking forward, we expect several trends to further develop ML in 2D FEs. Growing open databases and high-throughput computations will supply larger and more diverse training sets. Also, with more experience on applying ML to 2D FEs, using proper physics constraints (e.g. symmetry, conservation laws) will improve reliability. On the experimental side, combining ML with multiple data source (STM, X-ray, etc.) could yield better characterization. Importantly, a tight feedback loop between ML predictions and result verification will be crucial: models suggest candidates,

conventional methods test them, and the new data further refines the models.

In summary, ML in 2D ferroelectrics offers exceptional speed and acceptable accuracy, enabling rapid theoretical predictions and extrapolation of experiment results. But its practical effectiveness depends on careful handling of dataset and models. Balancing these factors, future work that combines physics-based understanding with data-driven methods will make ML a reliable and insightful tool for accelerating 2D ferroelectric research.

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